

8th Semester
CHEM-402
Past Paper
SHORT QUESTON
(2019)

a) Define soil and give its composition.

Soil refers to the upper layer of the Earth's crust that supports plant growth. It is a complex mixture of mineral particles, organic matter, water, air, and living organisms.

Composition:

Mineral Particles: *These include sand, silt, and clay. Sand particles are the largest, followed by silt, and then clay.*

Organic Matter: *Composed of decaying plant and animal material, providing nutrients and improving soil structure.*

Water: *Essential for plant growth and nutrient transport.*

Air: *Soil pores allow oxygen to reach plant roots.*

Living Organisms: *Bacteria, fungi, insects, earthworms, and plant roots contribute to soil health.*

b) What is meant by eutrophication? Give its main reasons.

Eutrophication *is the process by which a water body becomes enriched with nutrients (especially nitrogen and phosphorus), leading to excessive plant growth (algae and aquatic plants).*

Main Reasons:

Agricultural Runoff: *Excessive use of fertilizers in agriculture results in nutrient runoff into water bodies.*

Sewage Discharge: *Untreated sewage releases nutrients into rivers and lakes.*

Industrial Effluents: *Industries discharge nutrients into water systems.*

c) Differentiate between macronutrients and micronutrients giving



examples.

<u>Macronutrients:</u> Required in large quantities by plants. Examples: Nitrogen: Essential for growth, protein synthesis, and chlorophyll production. Phosphorus: Vital for energy transfer (ATP) and root development. Potassium: Regulates water balance and enzyme activity.	<u>Micronutrients:</u> Needed in small amounts but crucial for plant health. Examples: Iron: Involved in chlorophyll synthesis. Zinc: Aids enzyme function. Manganese: Necessary for photosynthesis.
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d) How is environment affected by PCBs?

Polychlorinated Biphenyls (PCBs):

Synthetic organic compounds used in electrical equipment, coolants, and lubricants.

Environmental Effects:

Persistence: PCBs remain in the environment for a long time.

Bioaccumulation: They accumulate in organisms, especially fish.

Toxicity: PCBs are harmful to wildlife and humans, affecting the nervous, immune, and reproductive systems.

e) How does pH of soil affect the availability of nutrients?

Soil pH affects nutrient availability to plants.

Acidic Soil (Low pH):

Aluminum Toxicity: High aluminum availability harms plant roots.

Reduced Nutrient Uptake: Phosphorus, calcium, and magnesium become less available.

Alkaline Soil (High pH):

Iron and Manganese Deficiency: These micronutrients are less accessible.

Reduced Phosphorus Availability: Phosphorus binds with calcium.

f) Give at least 4 volatile organic compounds that pollute the soil.

Four volatile organic compounds that pollute the soil include:



- 1) Benzene.
- 2) Toluene.
- 3) Xylene.
- 4) Trichloroethylene.

g) How is biodiversity affected by the presence of pollutants?

Biodiversity is affected by the presence of pollutants through various mechanisms such as habitat degradation, toxicity to organisms, disruption of ecological processes, and alteration of species composition. Pollutants can lead to the decline or loss of certain species, reduce genetic diversity, disrupt food chains and ecosystem functioning, and ultimately result in a decrease in overall biodiversity.

h) Define the term "ion-exchange capacity of soil".

The ion-exchange capacity of soil refers to its ability to retain and exchange ions with the surrounding soil solution. It is a measure of the soil's ability to adsorb and release cations (positively charged ions) and anions (negatively charged ions) from and to the soil solution, influencing nutrient availability and soil fertility.

i) How is environmental monitoring different than environmental analysis?

Environmental monitoring involves the systematic and ongoing collection, analysis, and interpretation of data to evaluate changes in environmental parameters over time. It focuses on monitoring environmental conditions, trends, and impacts to assess compliance with regulations, identify emerging issues, and guide management decisions.

Environmental analysis, on the other hand, typically refers to the laboratory-based measurement and quantification of specific pollutants or parameters in environmental samples to determine their concentration levels and potential effects.

j) How are micronutrients beneficial for plant growth?

Micronutrients are essential elements required by plants in small quantities for various physiological processes, including enzyme activation, photosynthesis, and growth regulation. They play crucial roles in plant metabolism, nutrient uptake, and defense mechanisms. Micronutrients such as iron, zinc, manganese, copper, boron, and molybdenum are involved in vital biochemical reactions and contribute to healthy plant growth, improved crop yield, and resistance to stressors such as pests, diseases, and environmental fluctuations.



(2020)

a) Toxicity of Cr is attributed to Cr (IV) not to the Cr (III). Why?

The toxicity of Cr is attributed to Cr(IV) rather than Cr(III) because Cr(VI) compounds are highly soluble and mobile in water, making them more bioavailable and easily absorbed by organisms. Once inside the body, Cr(VI) can undergo reduction to Cr(III), which can react with cellular components and DNA, leading to oxidative damage, genotoxicity, and carcinogenic effects.

b) What is CEC? Why soil can hold on cations.

*CEC stands for **Cation Exchange Capacity**, which is the ability of soil to retain and exchange cations (positively charged ions) with the soil solution. Soil can hold onto cations due to the presence of negatively charged exchange sites on clay minerals, organic matter, and soil colloids. These exchange sites attract and retain cations through electrostatic forces, allowing soil to act as a reservoir of essential nutrients for plant uptake.*

c) What are major causes of soil erosion?

Major causes of soil erosion include:

Water erosion: *Caused by rainfall, runoff, and flowing water, leading to detachment and transport of soil particles.*

Wind erosion: *Caused by wind action, resulting in the lifting and movement of soil particles.*

Tillage erosion: *Caused by agricultural activities such as plowing, cultivation, and land grading, disrupting soil structure and increasing susceptibility to erosion.*

d) What are aflatoxins?

Aflatoxins *are toxic metabolites produced by certain molds, particularly *Aspergillus* species that contaminate food crops such as maize, peanuts, and tree nuts. Aflatoxins are potent carcinogens and can cause liver damage, immune suppression, and growth impairment in humans and animals when ingested at high levels.*



e) Discuss control measures for water erosion?

Control measures for water erosion include:

- **Conservation tillage:** Minimizing soil disturbance and maintaining crop residues to protect soil surface and reduce runoff.
- **Terracing:** Constructing terraces or embankments on slopes to slow down water flow and trap sediment.
- **Cover crops:** Planting vegetation, such as grasses or legumes, to provide ground cover and stabilize soil.
- **Riparian buffers:** Establishing vegetated buffer zones along water bodies to filter sediment and reduce erosion.

f) Describe principle of UV-visible spectroscopy?

The principle of UV-visible spectroscopy involves the absorption of ultraviolet (UV) and visible light by molecules, resulting in electronic transitions between energy levels within the molecule. UV-visible spectroscopy measures the absorbance of light at specific wavelengths to characterize the electronic structure and concentration of analytes in solution. The amount of light absorbed is related to the concentration of the analyte and follows Beer's Law, allowing for quantitative analysis of substances in solution.

g) Discuss the main sources of lead in environment?

Lead-based paints: Historical use of lead-based paints in buildings, structures, and consumer products.

Lead-acid batteries: Disposal of lead-acid batteries from vehicles, electronics, and industrial equipment.

Industrial emissions: Release of lead particles from smelting, mining, metal processing, and combustion processes.

h) Why PCBs were banned?

PCBs (Polychlorinated Biphenyls) were banned due to their persistence, toxicity, and environmental impact. PCBs are synthetic organic compounds used in various industrial applications such as electrical equipment, hydraulic fluids, and plasticizers. They are persistent in the environment, bio-accumulate in the food chain, and pose significant health risks to humans and wildlife, including carcinogenic, immune-toxic, and reproductive effects.

i) What is atom economy?



Atom economy is a measure of the efficiency of a chemical reaction based on the proportion of reactant atoms that contribute to the desired product(s) compared to the total number of reactant atoms. It is calculated as the molecular weight of desired product(s) divided by the sum of the molecular weights of all reactants, expressed as a percentage. Atom economy encourages the design of synthetic routes that minimize waste generation, maximize resource utilization, and reduce environmental impact by optimizing the conversion of reactants into products.

(2022)

a) What is Humus?

Humus is the organic component of soil, formed by the decomposition of plant and animal matter. It is dark in color and rich in nutrients, playing a crucial role in soil fertility and structure.

b) How soil air is important for fertility?

Soil air is vital for fertility because it facilitates the exchange of gases necessary for plant growth, such as oxygen and carbon dioxide. Adequate soil air ensures proper root respiration and microbial activity, which in turn supports nutrient cycling and overall soil health.

c) Suggest some control measures for Water erosion?

Some control measures for water erosion include implementing contour plowing, terracing, planting cover crops, establishing buffer strips along water bodies, and constructing check dams or retention ponds to slow down water flow and trap sediment.

d) What type of pollutant can be analyzed with AAS?

Atomic Absorption Spectroscopy (AAS) is commonly used to analyze trace metals and metalloids in various environmental samples, including pollutants such as heavy metals in soil, water, and air.

e) How availability of Phosphorous is affected by soil pH?

The availability of phosphorus in soil is affected by soil pH because phosphorus tends to form insoluble compounds in alkaline soils, making it less available to plants. In acidic soils, phosphorus is more readily available for plant uptake.

f) What is chemical speciation?



Chemical speciation refers to the process of identifying and quantifying the different chemical forms or species of an element in a sample. This is important in environmental and analytical chemistry, as different chemical forms of an element may have different toxicities, bio-availabilities, or environmental fates.

g) Propose the reclamation of Basic soil?

Reclamation of basic soil can involve adding acidic materials such as sulfur or acidic fertilizers to lower the pH, incorporating organic matter to improve soil structure and fertility, and implementing erosion control measures to prevent further degradation.

h) Mention salient features of HPLC?

High-Performance Liquid Chromatography (HPLC) is a versatile analytical technique used for separating, identifying, and quantifying components in a mixture. Its salient features include high resolution, sensitivity, and the ability to analyze a wide range of compounds, making it suitable for various applications in fields like pharmaceuticals, environmental analysis, and food chemistry.

i) What are Primary and Secondary minerals of soil?

Primary minerals of soil are those originally formed during the soil formation process, such as quartz, feldspar, and mica. Secondary minerals are formed through weathering and alteration of primary minerals over time, such as clay minerals and iron oxides.

j) What are limitations of GC?

- *It is limited to gaseous sample only.*
- *Gas Chromatography (GC) include its inability to analyze non-volatile or thermally unstable compounds, the need for compound derivatization in some cases, and challenges in separating compounds with similar chemical properties.*

k) What is Dirty Zone?

The "Dirty Zone" refers to the interface between the Earth's crust and its atmosphere where various physical, chemical, and biological processes occur, leading to the formation and modification of soil.

l) Where is the word SOIL form?

The word "soil" is derived from the Latin word "solum," which means "earth" or "ground."

m) What is the importance of clay in nutrient holding capacity of soil?



Clay particles have a high surface area and a negative charge, allowing them to attract and hold onto positively charged nutrient ions such as potassium, calcium, and magnesium through cation exchange. This ability of clay to adsorb and retain nutrients contributes to the nutrient-holding capacity of soil, making them available for plant uptake over time.

CYANIDE

